FREE

The Ultimate Vitamin K2 Resource

If there’s a single vitamin you need to know more about, it’s vitamin K2.

CHRIS MASTERJOHN, PHD

APR 28, 2024

If there’s a single vitamin you need to know more about, it’s vitamin K_2. The first reason is you’re probably not getting enough. The second is that it doesn’t get the attention it deserves, and it’s really hard to find reliable and easy-to-use information about it.

This resource is meant to change that. It begins by teaching you everything you need to know about the vitamin, including its benefits, how much you need, and how to get it from food. It includes shareable infographics to make the concepts fun and easy to understand. Finally, it provides information on supplements and a searchable database of the vitamin K_2 contents of foods that can’t be found anywhere else.

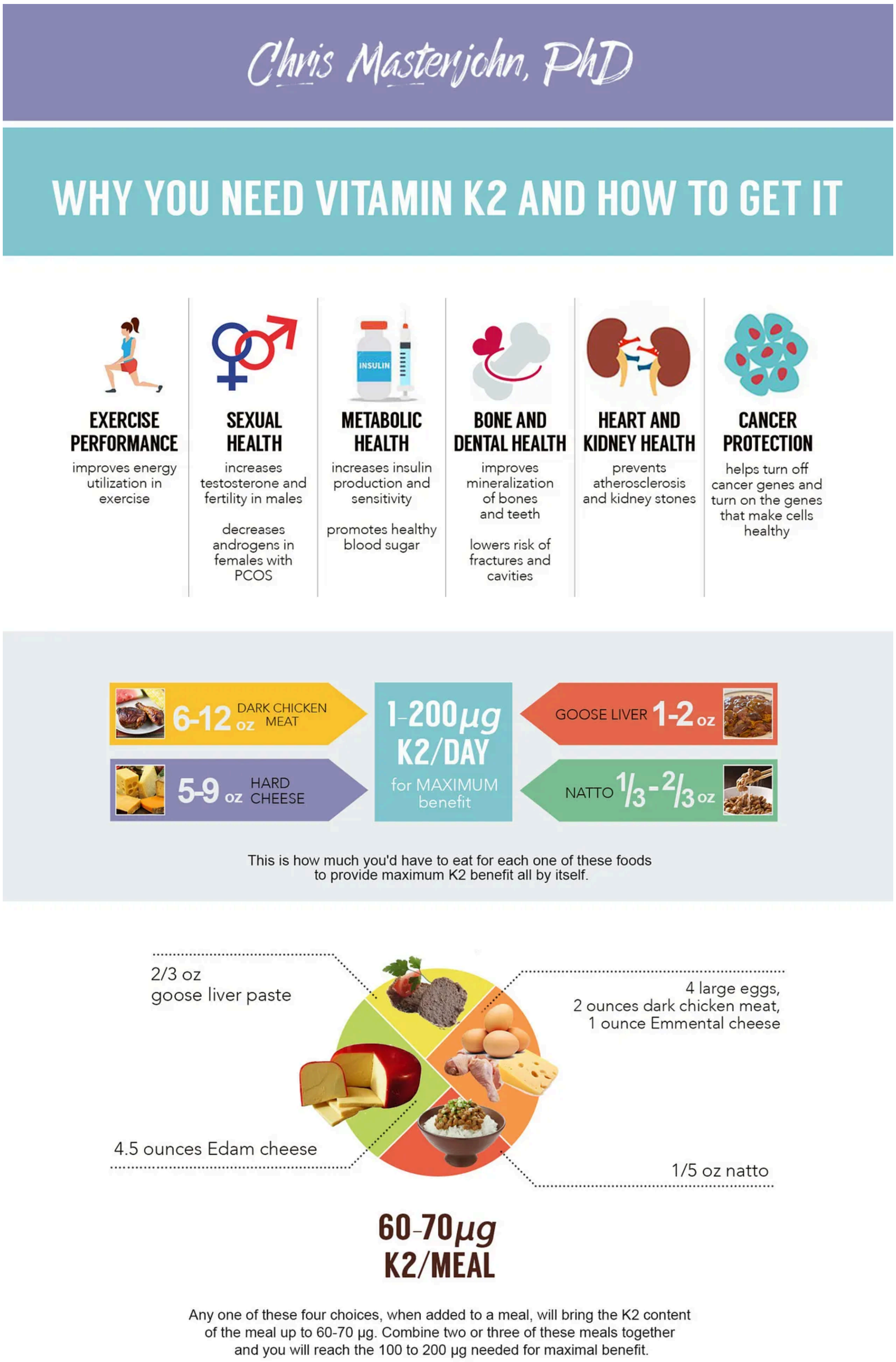
If you’re a beginner, you can read the article straight through or pick the parts that are most interesting or useful to you. If you are an advanced user and already know a lot about vitamin K_2 or have a strong science background, you can click on the buttons that say “click here for a more detailed explanation” in order to expand descriptions that are better suited to your level of expertise.

The Health Benefits of Vitamin K_2

Vitamin K_2 has a wide range of underappreciated health benefits:

- It prevents calcium from going into all the wrong places and makes sure it gets into all the right places. For example, it keeps it out of your kidneys, where it would cause kidney stones, and keeps it out of your blood vessels, where it would cause heart disease, but helps it to get into your bones and teeth, making your bones strong and your teeth resistant to cavities.
- It helps you make insulin and remain very sensitive to insulin. This means it helps stabilize your blood sugar, protects against diabetes, and prevents the metabolic problems that often arise as a consequence of obesity.
- It promotes sexual health by helping you optimize your sex hormones. For example, it increases testosterone and fertility in males, and it helps bring the high levels of male hormones found in women with polycystic ovarian syndrome (PCOS) back down to normal.
- It helps improve exercise performance by enhancing your ability to utilize energy during bouts of physical activity.
- It protects against cancer by suppressing the genes that make cells cancerous and expressing the genes that make cells healthy.

These roles are shown in the shareable infographic below. You can share it using the button in the upper right corner, or the buttons on the bottom strip. You can even use the button in the upper right corner to generate an embed code to share it on your own site if you have one.



Learn more about the biochemistry of vitamin K2 here:

The Biochemistry of Vitamin K2

Learn more about the evidence behind the health benefits of vitamin K2 here:

The Evidence for Vitamin K2

Why the Form of Vitamin K You Eat Is So Important

Vitamin K comes in different forms. Vitamin K_1 is primarily found in plant foods and is most abundant in leafy greens. Vitamin K_2 is only found in animal foods and fermented plant foods. The term "vitamin K_2 " actually refers to a collection of more specific forms known as menaquinones that are all abbreviated "MK" with a specific number attached: for example, MK-4, MK-7, MK-10, and so on.

Does it matter whether you eat one form or another? Absolutely. There are two reasons for this, so let's deal with them one at a time.

First, once we eat foods with vitamin K in them, our bodies handle the different forms differently. Consider these examples:

- **Vitamin K_1** travels to our livers more effectively than it does to our bones or blood vessels. The liver is where we use vitamin K to make the proteins involved in blood clotting, so vitamin K_1 is better at supporting blood clotting than it is at providing other health benefits.
- **MK-7** is much more effective than K_1 at reaching bone. This doesn't just make it good for bones: our bones use vitamin K to produce a hormone known as osteocalcin, which improves metabolic and hormonal health and increases exercise performance. Thus, MK-7 better supports these health benefits than K_1 . The portion of MK-7 that reaches the liver, moreover, stays active in the liver much longer than K_1 before being broken down; as a result, MK-7 is even better than K_1 at supporting blood clotting.
- **MK-4** is taken up by our tissues very rapidly after we consume it. While it hasn't been studied as carefully as MK-7, it may be less effective than MK-7 at reaching liver and bone but more effective at reaching most other tissues. This would make it better at protecting those tissues from calcium deposits and cancer development and supporting sex hormone production through its direct actions within our sex organs.

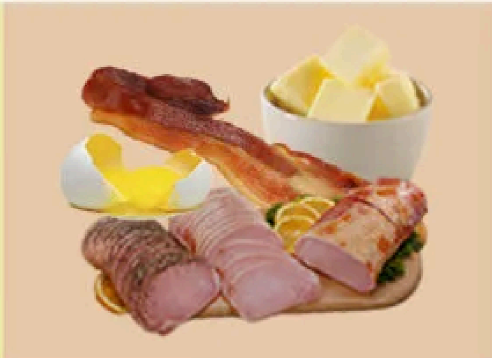
Overall, then, the collection of different vitamin K_2 compounds better supports all the health benefits listed above than vitamin K_1 because they better reach the tissues that matter.

These concepts are illustrated in the shareable infographic below.

Chris Masterjohn, PhD

WHY IT MATTERS WHAT TYPE OF VITAMIN K WE EAT

Vitamin K1 is most abundant in leafy greens.



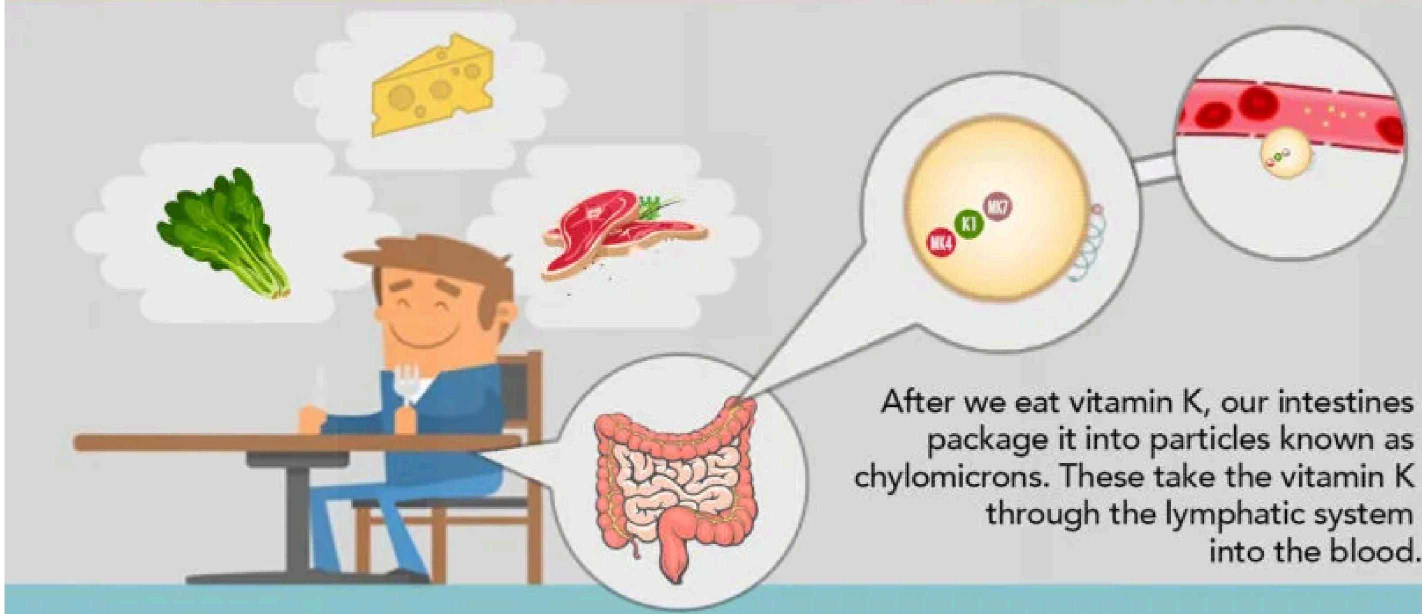
MK-4 is the form of vitamin K2 most abundant in animal foods.

MK-7, MK-8, and MK-9 are forms of vitamin K2 most abundant in fermented foods.

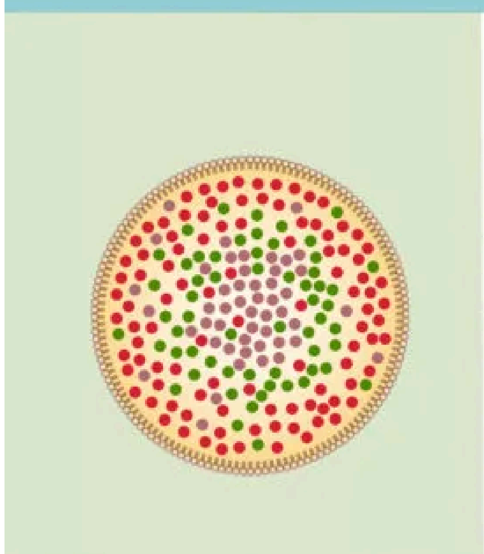
VITAMIN K1

VITAMIN K2

Vitamin K2 is a collection of different compounds known as MKs.

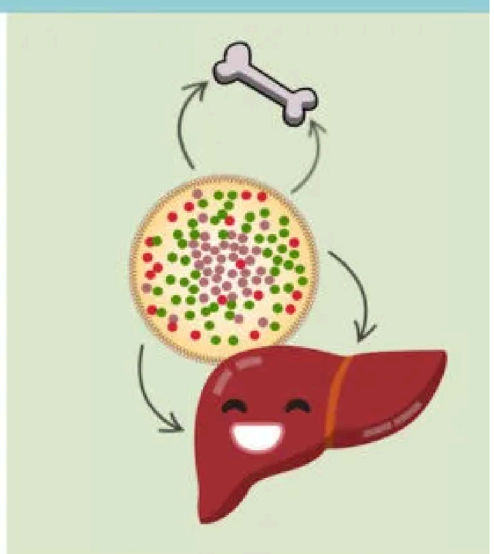


After we eat vitamin K, our intestines package it into particles known as chylomicrons. These take the vitamin K through the lymphatic system into the blood.

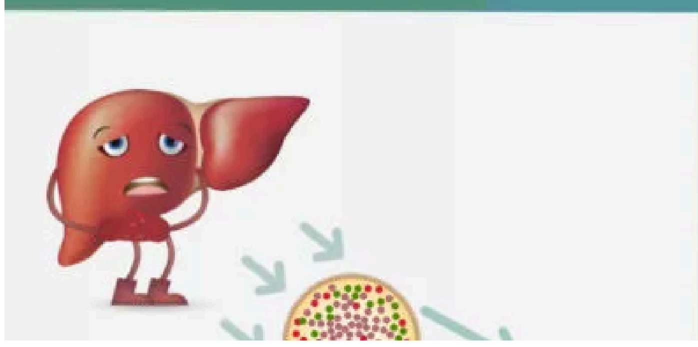


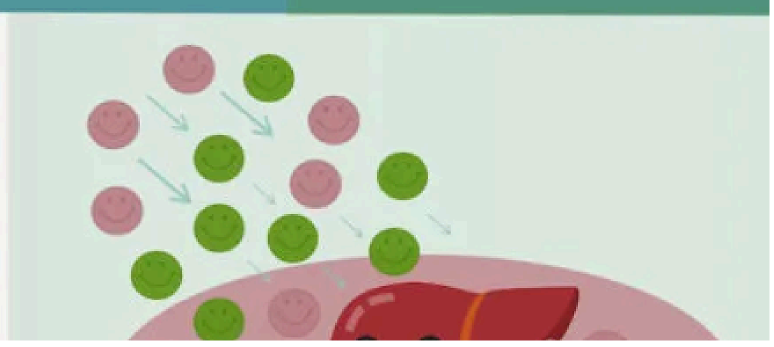
1 MK-4 is packed closer to the edges of the chylomicron, while MK-7 is packed in the center and K1 is packed in between them.

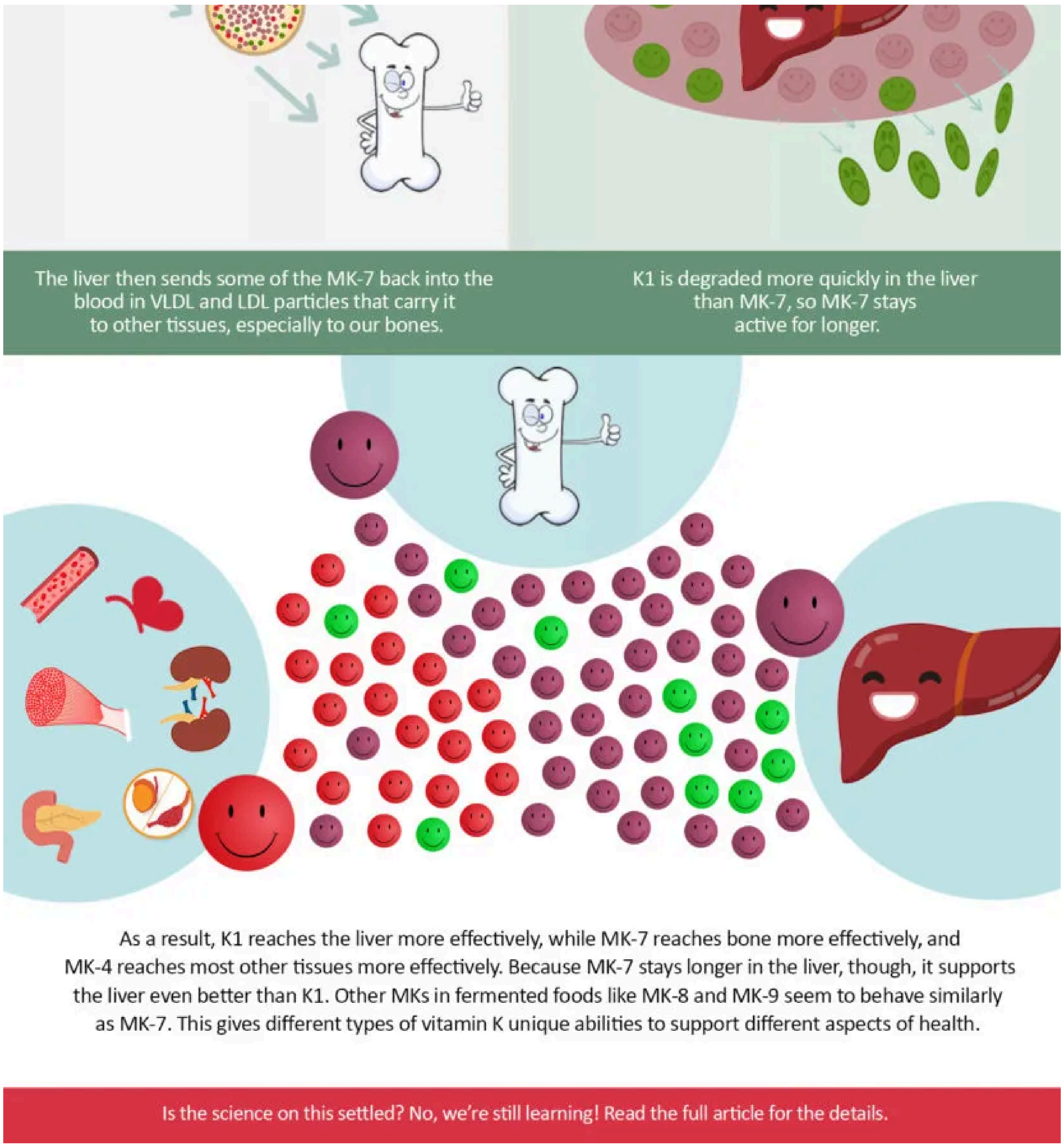
2 Most tissues bite off chunks of the chylomicron from the outside, where the MK-4 is.



3 Most of what's left over goes to the liver, while some goes to our bones.







The second reason the form of vitamin K matters is that MK-4 regulates gene expression in specific ways that no other form of vitamin K does. While we tend to think of our genes as the destiny we inherited from our parents, it's actually how they are expressed — meaning, what our cells do with the information carried by those genes — that determines our health. MK-4 turns on some genes and turns others off. For example, in our sex organs, it turns on the genes involved in sex hormone production. In a wide variety of cells, it turns on the genes that keep cells healthy and turns off the genes that make cells become cancerous. Thus, MK-4 plays an exclusive role in cancer protection and sexual health.

This special role of MK-4 probably explains why all animals break down other forms of vitamin K and convert them to MK-4. By contrast, no animal synthesizes any other form of vitamin K. This explains why MK-4 is mostly found in animal foods, and why most unfermented animal foods contain little if any of the other forms.

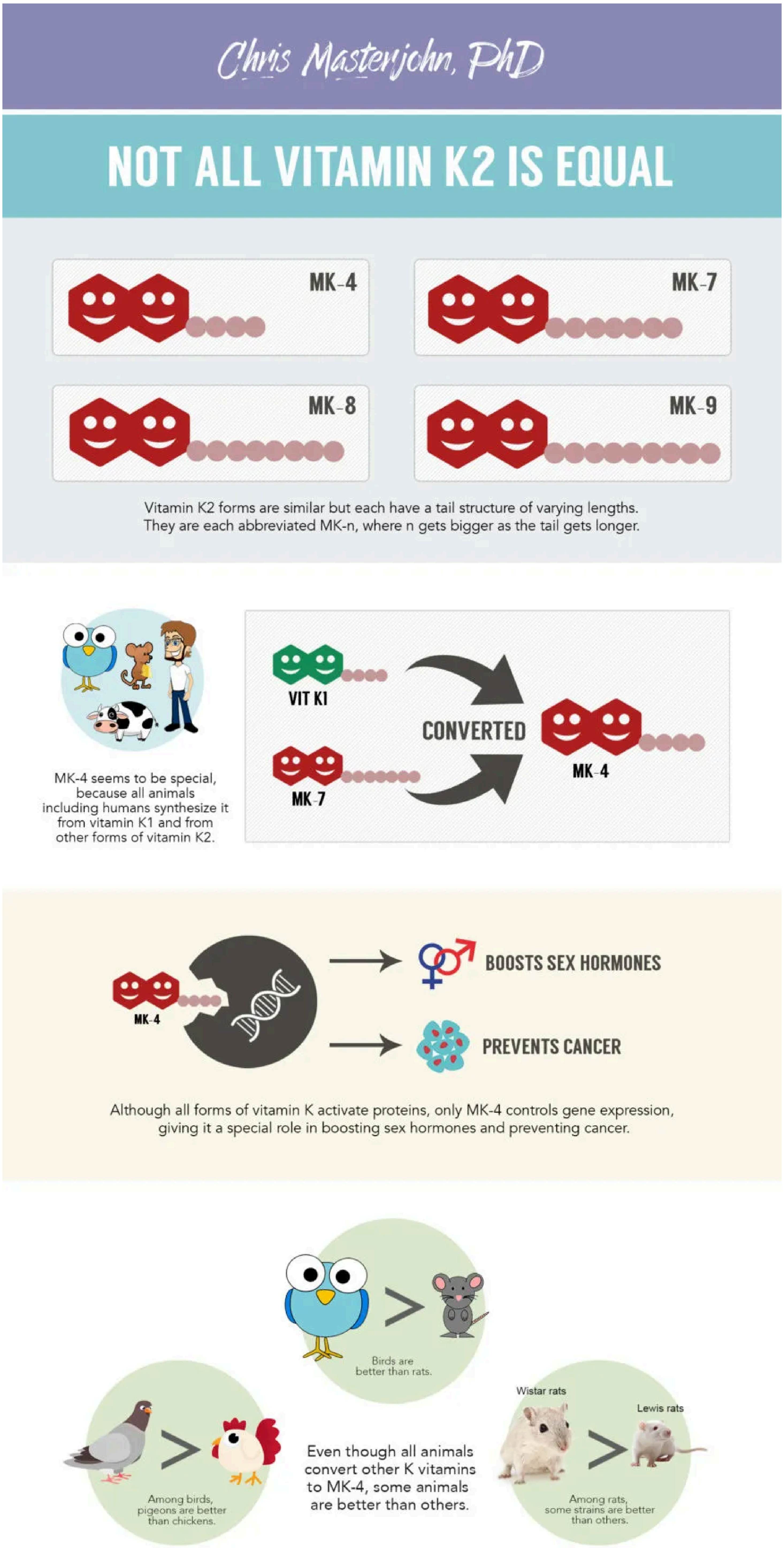
As humans, we also convert other forms of vitamin K to MK-4. This raises the question, do we really need to consume MK-4 directly if we can make it ourselves? My answer is yes.

There are three reasons we shouldn't rely on the conversion:

- First, we don't actually know that much about how the conversion takes place, but it seems to be inefficient and highly variable according to genetics and health status, making it unreliable.

- Second, cholesterol-lowering statin drugs and certain osteoporosis drugs inhibit the conversion, making it even less reliable in people who are taking these drugs.
- Third, research shows vitamin K_2 is better than vitamin K_1 at supporting many different aspects of our health. If we easily converted as much K_1 to K_2 as we needed, we wouldn't observe these superior benefits of K_2.

These concepts are illustrated in the shareable infographic below.


Chris Masterjohn, PhD
NOT ALL VITAMIN K2 IS EQUAL
MK-4
MK-7
MK-8
MK-9
Vitamin K2 forms are similar but each have a tail structure of varying lengths.
They are each abbreviated MK-n, where n gets bigger as the tail gets longer.
MK-4 seems to be special, because all animals including humans synthesize it from vitamin K1 and from other forms of vitamin K2.
VIT K1
MK-7
CONVERTED
MK-4
MK-4
BOOSTS SEX HORMONES
PREVENTS CANCER
Although all forms of vitamin K activate proteins, only MK-4 controls gene expression,
giving it a special role in boosting sex hormones and preventing cancer.
Birds are better than rats.
Among birds, pigeons are better than chickens.
Even though all animals convert other K vitamins to MK-4, some animals are better than others.
Wistar rats
Lewis rats
Among rats, some strains are better than others.

While a healthy diet should contain leafy greens (rich in K1) and fermented foods (rich in MK-7 through MK-10), consuming MK-4 from animal products may be important for maximum benefit.

The difference between K_1 and K_2 isn't absolute. When we eat vitamin K_1 some of it will reach tissues outside the liver and we will convert some of it to MK-4. But the real question is: what's the best vitamin for the job? Vitamin K_2 is clearly *much better* at supporting the health benefits discussed in this resource, so the resource is dedicated specifically to getting enough K_2 in its diversity of forms.

Learn more about the different forms of vitamin K here:

The Many Forms of Vitamin K

How Much Vitamin K_2 Do We Need?

Currently, there are no official recommendations about vitamin K_2. In the United States, the current recommendation for total vitamin K is 90 µg per day for adults. In a typical diet, most of this would come from K_2. These recommendations were last updated in 2001, before we learned about most of the benefits of K_2. In fact, the USDA did not even develop a database of vitamin K_2 in foods until 2006. My recommendation, therefore, does not rely on official sources and is meant for health-conscious people who wish to take advantage of cutting-edge research.

Based on the current state of that research, I recommend 100-200 µg per day of vitamin K_2 for healthy adults. Although most of the benefit probably comes from the first 100 µg, 200 µg is harmless and may provide additional benefit. If your health is fantastic while maintaining a K_2intake close to 100 µg, I would not worry about increasing your intake. But if you could stand to gain from the wide array of health benefits provided by the vitamin, I would use food or supplements to bring your intake closer to 200 µg.

Patients with chronic kidney disease may require doses as high as 480 μg per day and possibly much higher, but the use of high doses to treat a disease should always be done under medical supervision.

Patients using warfarin (Coumadin) or any other anticoagulant medications related to it should not make any changes to their vitamin K intake, regardless of the specific form of vitamin K, whether from food or supplements, except under the strict supervision of the prescribing physician (see below).

For more information, see here:

Vitamin K2: What Is the Optimal Dose?

How to Get Enough Vitamin K_2 From Food

You can use the searchable database I created to determine how much vitamin K_2 is in your diet. In this section, I describe a few of the simplest ways to get 200 μg per day of K_2 from foods. As noted above, most of the benefit comes from the first 100 μg, so any of the values below can be cut in half to obtain that amount.

The foods that are richest in K_2 are natto and goose liver, both of which may be difficult-to-acquire tastes. Natto is a fermented soy food popular in eastern Japan. The source of K_2 is the bacteria used in the fermentation, not the soy beans. As a result, any vegetable fermented with natto bacteria should be rich in K_2. For example, 100 grams of traditional natto contains just under 950 μg, while 100 grams of natto made from black beans contains almost 800 μg. The value for black bean natto is a little lower than that for traditional natto, but both values are phenomenally high. Simply adding 18 grams of natto (about two-thirds of an ounce) to your diet each day would give you 200 μg, and just two ounces of goose liver would provide the same benefit.

Another excellent source of vitamin K_2 is cheese. The K_2 content of cheese varies widely according to the type of bacteria used to make it. To browse a full list of cheeses, search "cheese" in our database or leave the search box blank and select the category "Dairy Foods and Eggs."

Jarlsberg cheese, which originates from Norway, is richest in K_2. According to the value listed in our database, it would take nine ounces of Jarlsberg to provide 200 μg. Its true content of K_2 has likely been underestimated, however, and it may actually take as little as 4.5 ounces.

Egg yolks and the dark meat (legs and thighs) of chicken are also good sources. For example, four whole eggs provides over 20 μg and 100 grams of dark chicken meat provides 60 μg.

Ultimately, it is the way these foods are combined in your diet that determines how much K_2 you get. The first infographic provides some ideas of how to work these different foods into a meal to make a meaningful contribution to your daily K_2 intake. You can figure out how much K_2 other meals would provide by using our database.

Surprisingly, we recently learned that pork products are very high in MK-10 and MK-11. This is a newly discovered exception to the rule that fresh animal products mostly contain MK-4. Unfortunately, little is known about the bioavailability of these forms and there are some indications that we as humans largely store them in our livers rather than distributing them throughout our bodies like other forms of K_2. However, if future research were to show that MK-10 and MK-11 have similar benefits as the other forms, that would mean most pork products are competitive sources. For example, only 4.5 ounces of baby back pork ribs would be needed to provide 200 µg, and just two ounces of pork sausage would provide the same amount.

Food quality is important. Egg yolk from The Netherlands is reported to have twice as much K_2 as egg yolk from the United States. The reasons for this are unclear, but it may relate to the ways the chickens were raised. Wherever possible, I recommend using meat, eggs, and dairy from animals raised on pasture. For egg yolks, look for the most deeply colored yolks you can find.

For more information, see here:

Vitamin K2 in Foods: A Closer Look

The Three Best Vitamin K_2 Supplements

Supplements should never be used to replace a good diet. A well-rounded nutrient-dense diet not only provides vitamin K itself in a greater diversity of forms than can be found in any supplement, but it also provides a full spectrum of other nutrients that work together with vitamin K to produce good health. As such, a good diet provides the context needed for a supplement to be both safe and effective.

When evaluating K_2 supplements, I look for the following things:

- **Dose: I prefer a dose from which it is easy to obtain approximately 200 ug. While it is probably effective to take a larger dose less than once a day (for example, taking 1 mg every five days), it is easier to maintain the habit of taking a daily dose.**

- **Form:** Since different forms of vitamin K_2 are distributed differently in the body, it is best to obtain a diversity of forms. In supplements, the best diversity we can obtain is to combine MK-4 and MK-7. The only supplemental MK-4 available is synthetic, but it is bioidentical, meaning it has the same chemical structure as the natural form. MK-7 on the market can be natural or synthetic; some synthetic MK-7 is bioidentical and some is not. Out of caution, I would only choose bioidentical options. Those who wish to have an entirely natural supplement should opt for MK-7 derived from the fermentation of soybeans or chickpeas.

- **Cost and convenience:** For any two products that are substantially equivalent, I prefer lower cost, easy online ordering, quick delivery, and the opportunity for free shipping.

Here are my top three recommendations:

- Innovix Labs Full Spectrum Vitamin K_2— This supplement wins on its diversity of forms (without going overboard on its total dose). It contains both MK-4 (500 µg) and MK-7 (100 µg). The MK-7 appears to be synthetic but bioidentical. It costs $21.97 on Amazon, is fulfilled by Amazon, and is eligible for Prime. Taken once a day, it costs 24 cents per day. Taken once every three days to achieve an average dose of 200 µg, it costs 9 cents per day.
- Thorne Research MK-4 — This supplement wins on cost. Its cost is nearly identical between Amazon and Iherb ($64.62 and $64.65), and if ordered on Amazon it is fulfilled by Amazon and eligible for Prime. It contains one milligram of MK-4 per drop. While the label recommends a daily dose of 45 drops, this is based on studies using pharmacological doses to treat osteoporosis. It is easy to instead take one drop per day to obtain a nutritional dose. Taken like that, it costs 5 cents per day. Taken once every five days to achieve an average dose of 200 µg, it costs one cent per day. They also make a combination of vitamin D and K_2 that is more expensive but easier to get a consistent daily dose of 200 µg from. This is described in more detail in the comprehensive review below.
- Nested Naturals K_2 — This is free of GMOs, soy, and other common allergens. It is made from fermented chickpeas. The MK-7 is made by another company, MenaQ7, whose MK-7 is sold under many different names and has also been used successfully in scientific research. This is the least expensive of all of the natural MK-7 products. It contains 100 µg of MK-7 and costs 12 cents per capsule. Taken twice per day to achieve 200 µg, it costs 24 cents per day.

An interesting runner-up is Nature's Plus, which is an affordable MK-7 supplement that is interesting mostly for its long list of features, like its background blend of plant, mushroom, and algae extracts, and its substantial list of third party certifications. It is described in more detail in the comprehensive review below.

If you have the time for home fermentation, Dr. Mercola created a starter culture that is designed to generate K_2 during the fermentation of vegetables. While I do not believe this will provide a standardized amount of K_2 like a commercial supplement will, I would expect it to substantially augment the K_2 content of your diet.

These are my top recommendations from a much more extensive review of over twenty supplements. If you want more details, click below for the comprehensive review.

Light and Heat Stability, and Proper Storage of Vitamin K_2

Vitamin K is only slightly sensitive to heat, but is extremely sensitive to light. So much so that when we measure vitamin K in a laboratory we work under yellow lamps. In food oils exposed to daylight, 80 percent of the vitamin K disappears within two days. To make sure that your food and supplements retain their vitamin K content over time, keep them in the refrigerator, behind cabinets, or otherwise out of the light when not in use. If you keep them in plain daylight, they should be in amber glass or in opaque containers such as the white plastic used for most supplements.

How Much Fat to Eat With Vitamin K_2 and What Kind

Vitamin K is fat-soluble so fat helps us absorb it from foods and supplements. If your fat intake varies from meal to meal, it makes sense to eat your K_2-rich foods or take your K_2 supplements with the meal that contains the most fat. The optimal amount of fat to maximize absorption of K_2 from a single meal is probably about 35 grams. The true optimal amount of fat has not been precisely determined and may be higher than this, but I consider it adequate.

For the best effect, the fat should be low in polyunsaturated fatty acids, which means that butter and other animal fats, tropical oils, olive oil, avocado oil, macadamia nut oil, and the high-oleic varieties of sunflower and safflower oil would help the most. By contrast, soybean oil, canola oil, the regular varieties of sunflower and safflower oil, grape seed oil, and most other oils derived from nuts and seeds would help the least.

Notably, many of the foods richest in K_2 like cheese, meat, and egg yolks are themselves rich in fat. The total fat content of the meal is what is important, so the more natural fats within the foods, the less you have to add.

To learn more, see here:

Vitamin K Absorption and Fat: A Closer Look

How to Test Your Vitamin K_2 Status

Unfortunately, there are no useful tests for measuring vitamin K status that are available to the general public at this time. However, good tests are on the horizon. VitaK will be releasing innovative medical devices to allow health care practitioners to monitor vitamin K status in patients, and ImmunoDiagnostic Systems will be releasing a blood test for dp-ucMGP, a protein that circulates in the blood when blood vessels become deficient in vitamin K.

For more information, see here:

Tests for Vitamin K Status: A Closer Look

Is Vitamin K_2 Dangerous?

Very high doses of vitamin K_2 have proved remarkably safe in large clinical trials, but there are safety concerns for people taking prescription anticoagulants, and there are reasons to be cautious about high doses even for healthy people.

This is Critical If You Are Taking Prescription Anticoagulants

The most common anticoagulants used in medicine are warfarin and its relatives. As a class, they are known as 4-hydroxycoumarins. These go by a number of brand names, the most common of which is Coumadin. As a class, these drugs act as vitamin K antagonists, and it is absolutely critical that anyone taking them avoid making any changes to their diet or supplements that would be expected to change their vitamin K intake except under the strict supervision of the physician who prescribed the medication.

Hypothetical Side Effects of High Doses

Long-term use of 45 mg per day of MK-4 has not revealed any established toxicity syndrome or risk of serious side effects. This is 225 times the dose I recommend. Nevertheless, the biochemistry of vitamin K suggests that unnecessarily high doses could rob the body of antioxidants or interfere with blood sugar regulation, insulin sensitivity, and hormonal health. The real question, though, is at what dose these potential side effects kick in. Since 45 mg per day has not shown any clear syndrome of toxicity and the dose I recommend is more than 200 times lower than this, I think we have a very large window of safety to work within. The potential for hypothetical side effects, however, should lead us to avoid supplementing with doses that are much larger than those that provide clear benefits.

For more information, see here:

[Hypothetical Side Effects of High-Dose Vitamin K]

Vitamin K_2: A Critical Component of a Well Rounded Nutrient-Dense Diet

Vitamin K_2 is something most of us could use a lot more of. The best way to obtain it is to consume K_2-rich foods in the context of a well rounded, nutrient-dense diet. The many other nutrients contained in a good diet provide the context that makes vitamin K_2 safe and effective. Supplements can be very helpful, as long as they are used as adjuncts to support a good diet rather than as replacements for a good diet.

Suggestions for Further Reading

In spring of 2007, I wrote "On the Trail of the Elusive X Factor: Vitamin K_2 Revealed." This is an extensive article arguing that vitamin K_2 was the "activator X" that Weston Price claimed to have discovered in 1945. Weston Price was one of the pioneers of nutritional anthropology, and many people had speculated about the identity of his "activator X" for decades. The article tells the history of that mystery and in the process extensively reviews the many roles of vitamin K and its interactions with other important nutrients like vitamins A and D.

For more about how vitamin K interacts with other nutrients in the diet, see my 2013 article, "Nutritional Adjuncts to the Fat-Soluble Vitamins."

For my other writings on vitamin K_2, see Start Here for Vitamin K_2.

Some other sources that I recommend include Chris Kresser's Vitamin K_2: The Missing Nutrient, and Stephan Guyenet's writings on the topic. Kate Rheaume-Bleue wrote a great book, Vitamin K_2 and the Calcium Paradox: How a Little-Known Vitamin Could Save Your Life.

For a more advanced understanding of vitamin K, I would start with the vitamin K chapter by John Suttie in *Modern Nutrition in Health and Disease*. I consider this textbook so valuable as a general scientific reference that I buy it again every time there is a new edition. Reviews by important figures in the field such as Martin Shearer, John Suttie, Sarah Booth, Cees Vermeer, and Leon Schurgers are also highly valuable and can be found on pubmed. On the specific topic of the hormonal functions of osteocalcin, I recommend reviews by Gerard Karsenty, also found on pubmed. Finally, the expandable "detailed explanation" sections within this resource are rich in scientific references that provide additional opportunities for advanced learning.

The Searchable Database of Vitamin K2 in Foods

The database is located here.

Comments

Over a thousand old comments can be found here. New comments can be read and posted below.

Discussion about this post

Comments Restacks

Write a comment...

JBR13 Apr 28, 2024 *Edited*

Masterjohn is a nutritional research machine - thank you for sharing the fruits of your labors!

LIKE (1) REPLY SHARE

JBR13 May 26 *Edited*

I tried to find oxalate content of natto, but there were no studies or clear data on that particular soybean product. Using a bunch of soybean cultivar data and studies showing oxalate reduction via soaking, cooking, and fermentation of soybeans and other legumes with B. subtilis, I conservatively, and very roughly, estimated the following regarding total (insoluble + soluble) oxalate content in natto:

Soybeans, on average, have 3.5-5mg total oxalate per gram soybean

Oxalate content is cut by 50% after: soaking, discarding the soaked water, boiling for 2-3 hours, and discarding the boil water.

Oxalate content decreases by 8% (of original raw soybeans) per day of fermentation at 100-105 degrees F. 8% of 3.5-5mg is 0.28-0.4mg (per gram of soybean).

1 tablespoon of natto weighs about 15g, and contains roughly 154mcg K2.

Total oxalate content of 1 tablespoon of 24hr-ferment natto is roughly and conservatively 24-38mg. Total oxalate should reduce by 0.28-0.4mg for each additional 24hr of fermentation at 100-105 degrees F.

...

My more liberal estimate using one of the lowest-oxalate cultivars of soybean is 8mg total oxalate in 1 tablespoon of 24hr-ferment natto.

My most conservative estimate using the highest-oxalate cultivar of soybean is 45mg total oxalate in 1 tablespoon of 24hr-ferment natto.

Average of 8mg, 24mg, 38mg, and 45mg is about 29mg total oxalate in 1 tablespoon natto

...

I'm also emailing companies that make natto to see if they have their own lab data on oxalate content. Boiled lentils - according to Harvard's oxalate data - is low-oxalate. Could use a natto starter and do an experiment like this - https://www.reddit.com/r/fermentation/comments/ueqdde/great_success_with_brown_lentil_natto/

LIKE REPLY SHARE

26 more comments...